The Case for Voluntary Disclosure

Strategic Feature Revelation in Regression and Classification

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Outline

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Motivation: Financial Firm Valuation

The Core Question

Should a predictive model enforce **mandatory** feature disclosure or allow **voluntary** disclosure by participants?

- **Scenario:** Market estimating a firm's valuation.
- **Issue:** Firms strategically reveal features. Some seek overvaluation (to raise stock prices), while others prefer undervaluation (e.g., for stock buybacks).
- **Intuition:** If a firm's preference aligns with the market objective, selective disclosure filters out noisy signals. If misaligned, it distorts information.

Sender-Receiver Game Model

The Setting:

- State drawn by Nature: x (features), y (ground truth), u (sender's preferred value).
- **Sender** observes (u, x) and chooses action $A \in \{\emptyset, x\}$ (withhold or disclose).
- **Noisy Channel:** Message M is received. If Sender discloses x, Receiver gets it with probability $1 - q$, and gets $\emptyset$ with probability q.

The Game:

- **Receiver** commits to strategy $\hat{y}(M)$.
- **Sender** perfectly best-responds by minimizing their expected loss $\ell(u, \hat{y}(M))$.
- **Stackelberg Equilibrium:** Receiver optimizes $\hat{y}$ anticipating Sender's rational reaction.

Mandatory vs. Voluntary Regimes

Mandatory Disclosure (Vanilla Risk)

Sender is forced to reveal $A = x$. Vanilla Risk: $R_V := \mathbb{E}[\ell(Y, \hat{y}_V(M))]$

Voluntary Disclosure (Strategic Risk)

Sender is allowed to withhold features opportunistically. Strategic Risk: $R_S := \min_{\hat{y}} \mathbb{E}[\ell(Y, \hat{y}(M))]$

Strategic Advantage

A **strategic advantage** exists if permitting voluntary disclosure yields a strictly lower expected loss for the Receiver: $R_S < R_V$.

Regression: Problem Setup

- **Outcome Space:** $\mathcal{Y} = \mathbb{R}$.
- **Loss Function:** Mean Squared Error $\ell(y, \hat{y}) = (y - \hat{y})^2$.
- All random variables (U, X, Y) are zero-mean.

Vanilla Risk (Mandatory Disclosure)

For a linear strategy parameterized by b (i.e., $\hat{y}_V(x) = b^\top x$ and $\hat{y}(\emptyset) = 0$), the optimal vanilla risk is achieved at $b_Y = \mathrm{Var}(X)^{-1}\,\mathrm{Cov}(X, Y)$:

$$R_V^\star(q) = \mathrm{Var}(Y) - (1 - q)\,\mathrm{Var}(b_Y^\top X)$$

Regression: Voluntary Disclosure Dynamics

Let the Receiver's linear strategy be parameterized by b.
We define two critical scalar fields:

$$F(x, u) := (2u - b^\top x)b^\top x \quad \text{(Sender's decision rule)}$$
$$G(x, y) := (2y - b^\top x)b^\top x \quad \text{(Receiver's relative gain)}$$

Sender's Best Response

The Sender discloses the feature $A(x, u) = x \iff F(x, u) > 0$.

Strategic Risk Formulation:

$$R_S(b, q) = \mathrm{Var}(Y) - (1 - q)\mathbb{E}\big[G(X, Y)\mathbb{I}\{F(X, U) > 0\}\big]$$

Regression: The Advantage Conditions

Theorem (Sufficiency Condition)

A strategic advantage ($R_S^\star(q) < R_V^\star(q)$) exists if there exists $b \in \mathbb{R}^d$ such that:

$$\|U - Y\|_2 < \frac{\mathbb{E}\big[|G(X,Y)|\big] - \mathrm{Var}(b_Y^\top X)}{4\sqrt{\mathrm{Var}(b_Y^\top X)}} - \frac{\mathrm{Var}((b - b_Y)^\top X)}{4\sqrt{\mathrm{Var}(b_Y^\top X)}}$$

Theorem (Necessary Condition)

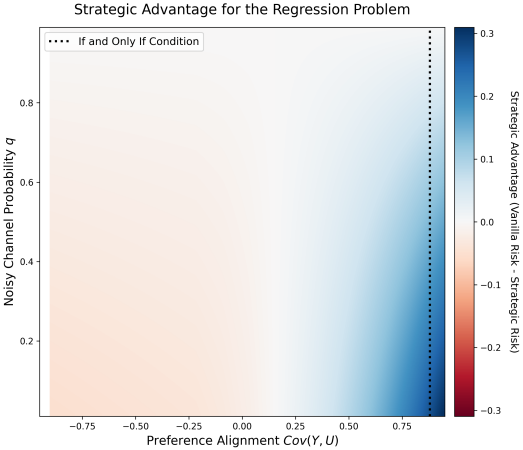
A necessary condition for a strategic advantage to exist is:

$$\max_b \Big(\mathbb{E}\big[|G(X,Y)|\big] - \mathrm{Var}(b_Y^\top X) - \mathrm{Var}((b - b_Y)^\top X)\Big) > 0$$

Insight: Both bounds are independent of channel noise q. Noise scales the magnitude but not the existence of the advantage.

Regression: Heatmap Verification

- Simulation over a zero-mean Gaussian family for (X, Y, U).
- $\mathrm{Cov}(X, Y) = \mathrm{Cov}(X, U) = 0.2$.
- Channel noise probability q swept along y-axis.
- Preference alignment $\mathrm{Cov}(Y, U)$ swept along x-axis.
- The dotted black line delineates theoretical "If and Only If" bounds!



Classification: Setup and Voluntary Responses

- **Outcome Space:** $\mathcal{Y} = \{-1, +1\}$ (Discrete Labels).
- **Loss Function:** $0-1$ Loss $\ell(y, \hat{y}) = \mathbb{I}\{\hat{y} \neq y\}$.
- **Vanilla Risk:** $R_V^\star(q) = q\mathbb{P}(Y = -\operatorname{sign}(\mathbb{E}[Y])) + (1-q)\mathbb{E}\left[\frac{1-|\eta(X)|}{2}\right]$

The Default Option Dynamics

Receiver commits to a **default prediction** $\hat{y}(\emptyset) = i \in \{-1, +1\}$.

- Sender with preference u withholds if $u = i$.
- Sender discloses if $u = -i$ to try shifting the prediction.
- Observing X thus reveals $U = -i$, prompting Receiver optimal posterior $\operatorname{sign}(\mathbb{E}[Y|X = x, U = -i])$.

Classification: Equilibrium and Advantage

Because there are only two default options, the Stackelberg Equilibrium simply chooses the default $\mathcal{S}_1$ or $\mathcal{S}_{-1}$ achieving minimal Strategic Risk.

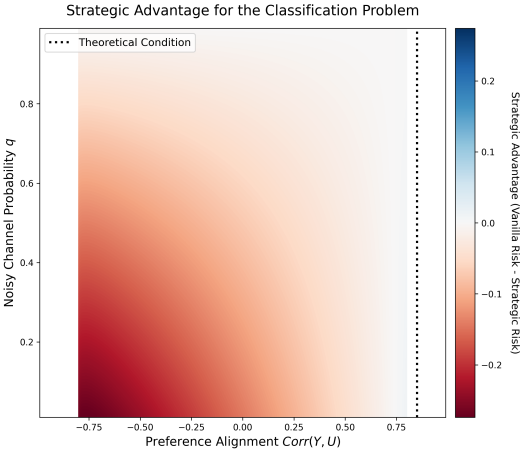
Theorem (Stackelberg Equilibrium)

- *For $q = 1$, strategy aligned with prior mean is strictly better.*
- *For lower q's, depending on distributions, an intermediate threshold $q^\star \in [0, 1]$ exists governing when the optimal default strategy switches.*

Unlike regression, channel noise q governs BOTH the **magnitude** *and* **direction** *of strategic advantage in specific distribution cases.*

Classification: Heatmap Verification

- Discrete labels $(Y, U) \in \{-1, 1\}^2$.
- Varying discrete preference alignment $\mathrm{Cov}(Y, U)$ on x-axis.
- Features from Gaussian Mixture Model $X|Y, U$.
- The theoretical boundary (dotted black) perfectly predicts advantage zones.



Learning in Unknown Environments

So far, we assumed the joint distribution (X, U, Y) was completely known. We now expand to:

1. **Static Learning:** Receiver estimates parameters and risks offline from a bounded population sample $\mathcal{D} = \{(x_i, u_i, y_i)\}_{i=1}^{n}$.
2. **Dynamic Learning:** Receiver learns parameters online across repeated interactions using Explore-Then-Commit strategies.

Static Learning: Generalization Bounds

Receiver computes empirical Vanilla Risk $\hat{R}_V$ and empirical Strategic Risk $\hat{R}_S$.

Theorem (Finite-Sample Generalization)

With probability $1-\delta$, estimation errors scale as $\mathcal{O}\left(\frac{1}{\sqrt{n}}\right)$:

$$|\hat{R}_V(q) - R_V(q)| \leq \mathcal{O}\left(\ldots\sqrt{\frac{d+\log(1/\delta)}{n}}\right)$$

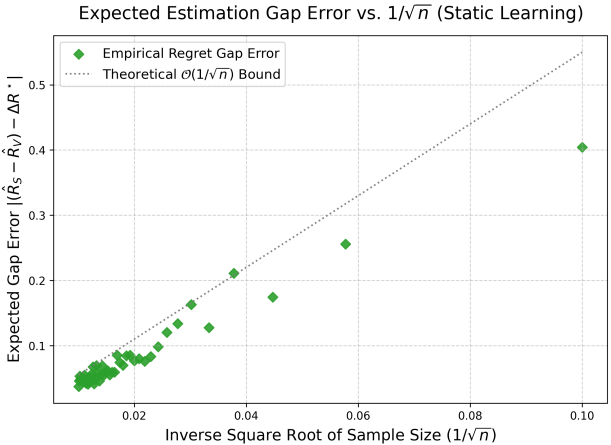
$$|\hat{R}_S(\hat{b}^\star, q) - R_S^\star(q)| \leq \mathcal{O}\left(\ldots\sqrt{\frac{d+\log(1/\delta)}{n}}\right)$$

Thus, the empirical regret gap reliably proxies the true Stackelberg equilibrium as $n \to \infty$.

Static Learning Empirical Convergence

Convergence Rate Simulation

- Expected gap error $|(\hat{R}_S - \hat{R}_V) - \Delta R^\star|$ shrinks linearly when plotted against $1/\sqrt{n}$.
- Confirms the $\mathcal{O}(1/\sqrt{n})$ theoretical reduction rate.



Dynamic Learning

Receiver minimizes cumulative regret over T sequential interactions:

$$\mathcal{R}_T = \sum_{t=1}^{T} \ell\big(y_t, \hat{y}_t(M_t)\big) - T \cdot \min\{R_V^\star, R_S^\star\}$$

Explore-Then-Commit (ETC) Algorithm

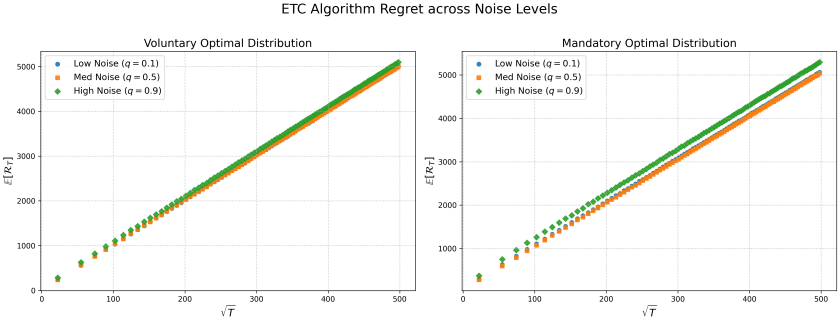
1. **Explore:** For $t = 1 \dots N$, enforce Voluntary regime. Observe data and maintain running empirical estimates $\hat{R}_V, \hat{R}_S$.
2. **Switch:** At $N + 1$, permanently commit to whichever regime (Voluntary or Mandatory) had lower estimated risk.
3. **Commit:** Continually refine the predictor weights online under the designated regime for the remainder of time T.

Dynamic Learning Regret Analysis

Theorem

The ETC Disclosure Algorithm achieves sublinear expected regret:

$$\mathbb{E}[\mathcal{R}_T] = \mathcal{O}(\sqrt{T})$$



Conclusion

- **Strategic Disclosures:** We modeled feature revelation as a game and showed that under certain quantifiable conditions, allowing participants to withhold features **improves** receiver's predictive accuracy.
- **Regression vs Classification:** Derived rigid "if and only if" bounds for Strategic Advantage, dependent on Sender-Receiver alignment $\mathrm{Cov}(U, Y)$.
- **Static and Dynamic Learning:** We showed that these concepts translate reliably to real-world learning problems where environment parameters must be deduced from samples or interactions, achieving $\mathcal{O}(\sqrt{T})$ dynamic regret limits.

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